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United States Patent	12385661
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Honda; Kazuya et al.

Air-conditioning apparatus

Abstract

An air-conditioning apparatus includes configured to, when a cooling operation mode is changed into one of multiple dehumidifying operation modes, select the one of the multiple dehumidifying operation modes on the basis of a value of a sensible heat ratio difference ΔSHF that is a difference between a target sensible heat ratio acquired from an indoor temperature detected by an indoor temperature sensor, a target indoor temperature, and target indoor humidity and a theoretical minimum possible sensible heat ratio acquired from enthalpy of an indoor air, enthalpy of blown air from the indoor unit when relative humidity is 100%, and enthalpy of sensible heat.

Inventors:	Honda; Kazuya (Tokyo, JP), Takenaka; Naofumi (Tokyo, JP), Takagi; Jumpei (Tokyo, JP), Tomita; Masafumi (Tokyo, JP), Sakai; Mizuo (Tokyo, JP)
Applicant:	Mitsubishi Electric Corporation (Tokyo, JP)
Family ID:	83322069
Assignee:	MITSUBISHI ELECTRIC CORPORATION (Tokyo, JP)
Appl. No.:	18/277583
Filed (or PCT Filed):	March 18, 2021
PCT No.:	PCT/JP2021/010974
PCT Pub. No.:	WO2022/195791
PCT Pub. Date:	September 22, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240133573 A1	Apr. 25, 2024
US 20240230137 A9	Jul. 11, 2024

Publication Classification

Int. Cl.: F24F11/65 (20180101); F24F1/0083 (20190101); F24F11/00 (20180101); F24F110/20 (20180101)

U.S. Cl.:

CPC F24F11/65 (20180101); F24F1/0083 (20190201); F24F11/0008 (20130101); F24F2110/20 (20180101)

Field of Classification Search

CPC: F24F (11/65); F24F (11/0008); F24F (11/46); F24F (11/86); F24F (11/30); F24F (11/64); F24F (11/74); F24F (1/0083); F24F (2110/20); F24F (2110/10); F24F (3/14)

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Primary Examiner: Nieves; Nelson J

Attorney, Agent or Firm: XSENSUS LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application is based on PCT filing PCT/JP2021/010974, filed Mar. 18, 2021, the

entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an air-conditioning apparatus that performs a dehumidifying operation.

BACKGROUND ART

(3) An air-conditioning apparatus controls the rotation frequency of a compressor such that an indoor temperature approaches a predetermined temperature during a cooling operation. As a temperature difference between the predetermined temperature and the indoor temperature decreases, the rotation frequency of the compressor decreases, and an evaporating temperature in an indoor heat exchanger becomes more than or equal to the dew point of indoor air. This poses a problem in that dehumidification is not performed and comfort is reduced. In view of this, during a cooling operation, a known air-conditioning apparatus sets the evaporating temperature of refrigerant at which a sensible heat ratio in the indoor heat exchanger is less than or equal to a predetermined reference value to the maximum possible evaporating temperature and controls the evaporating temperature of the refrigerant in the indoor heat exchanger within the range of the maximum possible evaporating temperature or less (see, for example, Patent Literature 1).

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Patent No. 5487857

SUMMARY OF INVENTION

Technical Problem

(5) In the air-conditioning apparatus disclosed in Patent Literature 1, however, the evaporating temperature is set on the basis of the sensible heat ratio in the indoor heat exchanger, and for this reason, the evaporating temperature may greatly decrease to ensure the amount of dehumidification in some cases. This poses a problem in that the indoor temperature greatly decreases and user comfort is reduced.

(6) The present disclosure has been accomplished to solve the above problems, and it is an object to provide an air-conditioning apparatus that can perform a dehumidifying operation without reducing user comfort.

Solution to Problem

(7) An air-conditioning apparatus according to an embodiment of the present disclosure includes an indoor unit including an indoor heat exchanger, an indoor fan configured to send indoor air to the indoor heat exchanger, and an indoor temperature sensor configured to detect an indoor temperature, the indoor unit being installed in an indoor space, an outdoor unit including a compressor and an outdoor heat exchanger, the outdoor unit being installed outside the indoor space, a refrigerant circuit in which the compressor, the outdoor heat exchanger, an expansion device, and the indoor heat exchanger are connected by a pipe, and a controller having a cooling operation mode in which the indoor space is cooled and multiple dehumidifying operation modes in which the indoor space is dehumidified. The multiple dehumidifying operation modes include a first dehumidifying operation mode in which an air volume of the indoor fan is increased or decreased and a second dehumidifying operation mode in which the compressor alternates between operation and suspension. The controller is configured to, when the cooling operation mode is changed into one of the multiple dehumidifying operation modes, select the one of the multiple dehumidifying operation modes on the basis of a value of a sensible heat ratio difference ΔSHF that is a difference between a target sensible heat ratio acquired from the indoor temperature detected by the indoor temperature sensor, a target indoor temperature, and target indoor humidity and a theoretical minimum possible sensible heat ratio acquired from enthalpy of the indoor air, enthalpy of blown air from the indoor unit when relative humidity is 100%, and enthalpy of sensible heat.

Advantageous Effects of Invention

(8) In the air-conditioning apparatus according to an embodiment of the present disclosure, the controller is configured to, when the cooling operation mode is changed into one of the multiple dehumidifying operation modes, select the one of the multiple dehumidifying operation modes having different dehumidifying capacities on the basis of the value of the sensible heat ratio difference ΔSHF , which is a difference between the target sensible heat ratio and the theoretical minimum possible sensible heat ratio. This enables a dehumidifying operation mode suited to dehumidification to be selected, prevents the evaporating temperature from greatly decreasing, and enables dehumidification to be performed without reducing user comfort.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a refrigerant circuit diagram illustrating an example of the structure of an air-conditioning apparatus according to Embodiment 1.
- (2) FIG. 2 is a flowchart for the air-conditioning apparatus according to Embodiment 1 when a cooling operation mode is changed into a dehumidifying operation mode.
- (3) FIG. 3 is a chart illustrating an example of a control operation of the air-conditioning apparatus according to Embodiment 1 in a first dehumidifying operation mode.
- (4) FIG. 4 is a chart illustrating an example of a control operation of the air-conditioning apparatus according to Embodiment 1 in a second dehumidifying operation mode.
- (5) FIG. 5 is a refrigerant circuit diagram illustrating an example of the structure of an air-conditioning apparatus according to Embodiment 2.
- (6) FIG. 6 is a flowchart for the air-conditioning apparatus according to Embodiment 2 when the cooling operation mode is changed into a dehumidifying operation mode.
- (7) FIG. 7 is a saturated air psychrometric chart illustrating an example of the state of air in the air-conditioning apparatus according to Embodiment 2 in a third dehumidifying operation mode.
- (8) FIG. 8 is a refrigerant circuit diagram illustrating an example of the structure of an air-conditioning apparatus according to Embodiment 3.
- (9) FIG. 9 is a flowchart for the air-conditioning apparatus according to Embodiment 3 when the cooling operation mode is changed into a dehumidifying operation mode.
- (10) FIG. 10 is a flowchart for an air-conditioning apparatus according to Embodiment 4 when the cooling operation mode is changed into a dehumidifying operation mode.

DESCRIPTION OF EMBODIMENTS

(11) Embodiments of the present disclosure will hereinafter be described with reference to the drawings. The present disclosure is not limited by the embodiments described below. The relationship between the sizes of components in some drawings described below may differ from the relationship between the sizes of actual ones in some cases.

Embodiment 1

(12) [Air-Conditioning Apparatus]

(13) FIG. 1 is a refrigerant circuit diagram illustrating an example of the structure of an air-conditioning apparatus **100** according to Embodiment 1.

(14) As illustrated in FIG. 1, the air-conditioning apparatus **100** according to Embodiment 1 includes an outdoor unit **1** and an indoor unit **3a**. The outdoor unit **1** and the indoor unit **3a** are connected to each other by a gas pipe **15** and a liquid pipe **16** and are included in a refrigerant circuit.

(15) [Outdoor Unit]

(16) The outdoor unit **1** is installed, for example, outside a room and removes or provides heat for air conditioning. For example, the outdoor unit **1** includes a compressor **11**, a flow switching device **14**, an outdoor heat exchanger **12**, and an expansion device **13**, which are connected by pipes. The

outdoor unit **1** includes an outdoor fan **121** that sends air to the outdoor heat exchanger **12**. The expansion device **13** may be included in the indoor unit **3a** instead of in the outdoor unit **1**. The compressor **11** sucks and compresses refrigerant into high-temperature and high-pressure refrigerant and is, for example, an inverter compressor, which has a controllable capacity. The flow switching device **14** switches between a refrigerant flow path for a cooling operation and a refrigerant flow path for a heating operation and changes a heat exchanger that serves as a condenser or a gas cooler. During the cooling operation, the flow switching device **14** is switched such that a discharge port of the compressor **11** is in communication with the outdoor heat exchanger **12** and such that the gas pipe **15** is in communication with a suction port of the compressor **11**. Consequently, the outdoor heat exchanger **12** serves as the condenser or the gas cooler. During the heating operation, the flow switching device **14** is switched such that the discharge port of the compressor **11** is in communication with the gas pipe **15** and such that the outdoor heat exchanger **12** is in communication with the suction port of the compressor **11**. Consequently, the outdoor heat exchanger **12** serves as an evaporator. The flow switching device **14** is a four-way valve but is not limited to a four-way valve. The flow switching device **14** may include a device such as a two-way valve.

(17) The outdoor heat exchanger **12** exchanges heat between air and the refrigerant and consequently condenses or evaporates the refrigerant. The outdoor fan **121** sends outdoor air to the outdoor heat exchanger **12**. The outdoor fan **121** is driven by an outdoor fan motor **122**. The expansion device **13** decompresses and expands the refrigerant and is, for example, an electronic expansion valve, which has an adjustable expansion opening degree.

(18) [Indoor Unit]

(19) The indoor unit **3a** is installed, for example, inside a room and supplies air for air conditioning to an indoor space. The indoor unit **3a** includes an indoor heat exchanger **31a**. The indoor unit **3a** also includes an indoor fan **33a** that sends air to the indoor heat exchanger **31a**.

(20) The indoor heat exchanger **31a** exchanges heat between air and the refrigerant, consequently condenses or evaporates the refrigerant, and generates heating air or cooling air to be supplied to the indoor space. The indoor fan **33a** sends indoor air to the indoor heat exchanger **31a**. The indoor fan **33a** is driven by an indoor fan motor **34a**.

(21) The indoor unit **3a** also includes an indoor temperature sensor **35a**, an entrance temperature sensor **36a**, and an exit temperature sensor **37a**. The indoor temperature sensor **35a** is, for example, a thermistor and detects an indoor temperature T_a . The indoor temperature sensor **35a** is disposed at an air inlet of the indoor unit **3a** for the indoor air. The entrance temperature sensor **36a** is, for example, a thermistor and detects the temperature of the refrigerant that flows into the indoor heat exchanger **31a** during the cooling operation. The entrance temperature sensor **36a** is disposed at the entrance of the indoor heat exchanger **31a** for the refrigerant during the cooling operation. The exit temperature sensor **37a** is, for example, a thermistor and detects the temperature of the refrigerant that flows out from the indoor heat exchanger **31a** during the cooling operation. The exit temperature sensor **37a** is disposed at the exit of the indoor heat exchanger **31a** for the refrigerant during the cooling operation.

(22) The refrigerant circuit of the air-conditioning apparatus **100** according to Embodiment 1 is formed by connecting the compressor **11**, the flow switching device **14**, the outdoor heat exchanger **12**, the expansion device **13**, and the indoor heat exchanger **31a** in this order by the pipes.

(23) The air-conditioning apparatus **100** also includes a controller **50**. The controller **50** controls, for example, the entire air-conditioning apparatus **100** and includes, for example, an analog circuit, a digital circuit, a CPU, or a combination of two of these. The controller **50** controls, for example, the driving frequency of the compressor **11**, turning on and off of the indoor fan **33a** and the outdoor fan **121**, the rotation frequencies of the indoor fan **33a** and the outdoor fan **121**, switching of the flow switching device **14**, and the opening degree of the expansion device **13**, for example, on the basis of information that is detected by the sensors described above and an instruction from

an input device such as a remote controller and activates operation modes described later. In FIG. 1, the outdoor unit 1 includes the controller 50, but this is not a limitation. The indoor unit 3a may include the controller 50, and the outdoor unit 1 and the indoor unit 3a may also include the respective controllers 50.

(24) [Operation Modes of Air-conditioning Apparatus]

(25) The operation modes in which the air-conditioning apparatus 100 operates will be described below.

(26) The air-conditioning apparatus 100 has a cooling operation mode and dehumidifying operation modes. The flow of the refrigerant and the state of the refrigerant in each operation mode will be described below.

(27) [Cooling Operation Mode] The cooling operation mode in which the indoor heat exchanger 31a of the air-conditioning apparatus 100 according to Embodiment 1 generates a cooling load will be described.

(28) The flow of the refrigerant will first be described. The compressor 11 sucks and compresses low-temperature and low-pressure gas refrigerant and discharges high-temperature and high-pressure gas refrigerant. The high-temperature and high-pressure gas refrigerant that is discharged from the compressor 11 flows into the outdoor heat exchanger 12 via the flow switching device 14. The outdoor heat exchanger 12 exchanges heat between the outdoor air that is supplied from the outdoor fan 121 and the high-temperature and high-pressure gas refrigerant. Intermediate-temperature and high-pressure refrigerant that is acquired by cooling the refrigerant at the outdoor heat exchanger 12 flows into the expansion device 13. The expansion device 13 decompresses the intermediate-temperature and high-pressure refrigerant in a two-phase or liquid state. Low-temperature and low-pressure two-phase refrigerant that is acquired by decompressing the refrigerant at the expansion device 13 flows into the indoor heat exchanger 31a via the liquid pipe 16. At this time, the indoor unit 3a performs the cooling operation, and the indoor heat exchanger 31a exchanges heat between the indoor air that is supplied from the indoor fan 33a and the low-temperature refrigerant. Intermediate-temperature refrigerant that is acquired by heating the refrigerant at the indoor heat exchanger 31a passes through the gas pipe 15 and the flow switching device 14 and is sucked by the compressor 11.

(29) The compressor 11 is controlled such that a temperature difference ΔT between a target indoor temperature T_m and the indoor temperature T_a detected by the indoor temperature sensor 35a approaches 0. For example, the larger the temperature difference ΔT , the higher the rotation frequency of the compressor 11, and the smaller the temperature difference ΔT , the lower the rotation frequency of the compressor 11. An air volume from the indoor unit 3a is controlled by the controller 50. The controller 50 controls the air volume from the indoor unit 3a within an adjustable range. For example, the upper limit of the adjustable range is 100% of the maximum possible air volume of the indoor fan 33a, and the lower limit of the adjustable range is 70% of the maximum possible air volume.

(30) [Conditions for Change into Dehumidifying Operation Modes]

(31) Conditions for change from the cooling operation mode into the dehumidifying operation modes will be described below. When the temperature difference ΔT described above becomes a predetermined temperature or less, and the amount of change in the rotation frequency of the compressor 11 becomes a predetermined value or less, whether a change into one of the dehumidifying operation modes is made is determined.

(32) [Determination of Dehumidifying Operation Mode]

(33) FIG. 2 is a flowchart for the air-conditioning apparatus 100 according to Embodiment 1 when the cooling operation mode is changed into one of the dehumidifying operation modes.

(34) The dehumidifying operation modes of the air-conditioning apparatus 100 according to Embodiment 1 include a first dehumidifying operation mode and a second dehumidifying operation mode.

(35) The flow of change from the cooling operation mode into one of the dehumidifying operation modes of the air-conditioning apparatus **100** according to Embodiment 1 will be described below with reference to FIG. 2.

(36) (Step S1)

(37) The controller **50** acquires the indoor temperature T_a detected by the indoor temperature sensor **35a**. Subsequently, processing proceeds to step S2.

(38) (Step S2)

(39) The controller **50** acquires a target sensible heat ratio SHF from the acquired indoor temperature T_a , the target indoor temperature T_m , and target indoor humidity. Subsequently, the processing proceeds to step S3.

(40) (Step S3)

(41) The controller **50** acquires a sensible heat ratio difference ΔSHF , which is a difference between the target sensible heat ratio SHF and the theoretical minimum possible sensible heat ratio SHF_m . Specifically, the sensible heat ratio difference ΔSHF is acquired by subtracting the theoretical minimum possible sensible heat ratio SHF_m from the target sensible heat ratio SHF. Subsequently, the processing proceeds to step S4. The theoretical minimum possible sensible heat ratio SHF_m is acquired by the following expression.

$$SHF_m = \Delta h_s / (h_i - h_o) \quad [\text{Math. 1}]$$

(42) Here, h_i is the enthalpy of the indoor air that is acquired from the indoor temperature and relative humidity, h_o is the enthalpy of air (referred to below as blown air) that is blown via an air outlet of the indoor unit **3a** when the relative humidity is 100%, and Δh_s is the enthalpy of sensible heat. The enthalpy of the sensible heat can be calculated by $\Delta h_s = (T_{in} - T_{out}) \times C_p$, where T_{in} is the temperature of air that is sucked via the air inlet of the indoor unit **3a**, T_{out} is the temperature of the air that is blown via the air outlet of the indoor unit **3a**, and C_p is the specific heat of air. The smaller the value of h_i , the higher the theoretical minimum possible sensible heat ratio SHF_m , and the larger the value of h_o , the higher the theoretical minimum possible sensible heat ratio SHF_m . In view of this, for example, the temperature of the blown air regarding h_o may be extremely lower than the temperature of the indoor air regarding h_i to decrease the theoretical minimum possible sensible heat ratio SHF_m and to increase the sensible heat ratio difference ΔSHF .

(43) (Step S4)

(44) The controller **50** determines whether the sensible heat ratio difference ΔSHF is larger than a reference value $X1$. When the controller **50** determines that the sensible heat ratio difference ΔSHF is larger than the reference value $X1$ (YES), the processing proceeds to step S5. On the other hand, when the controller **50** determines that the sensible heat ratio difference ΔSHF is not larger than the reference value $X1$ (NO), the processing proceeds to step S6. The reference value $X1$ is, for example, 0.2 to 0.3, is determined depending on the dew point of the indoor air, and increases as the dew point of the indoor air decreases.

(45) (Step S5)

(46) The controller **50** changes the operation mode from the cooling operation mode into the first dehumidifying operation mode. Subsequently, the processing returns to step S1.

(47) (Step S6)

(48) The controller **50** changes the operation mode from the cooling operation mode into the second dehumidifying operation mode. Subsequently, the processing returns to step S1.

(49) [First Dehumidifying Operation Mode]

(50) FIG. 3 is a chart illustrating an example of a control operation of the air-conditioning apparatus **100** according to Embodiment 1 in the first dehumidifying operation mode. In FIG. 3, the horizontal axis represents the sensible heat ratio difference, and the vertical axis represents the air volume of the indoor fan **33a**. Arrows in FIG. 3 represent transition of the air volume.

(51) The first dehumidifying operation mode of the air-conditioning apparatus **100** according to Embodiment 1 will be described below with reference to FIG. 3.

(52) In the first dehumidifying operation mode, the controller **50** controls the air volume of the indoor fan **33a**. The lower limit of the air volume of the indoor fan **33a** may be less than the lower limit of the air volume in the cooling operation mode. As illustrated in FIG. 3, the air volume of the indoor fan **33a** is controlled depending on the magnitude of the sensible heat ratio difference ΔSHF . In FIG. 3, the air volume of the indoor fan **33a** is controlled such that the air volume is set to one of three air volumes 1 to 3 (the air volume 1 > the air volume 2 > the air volume 3). However, this is not a limitation. The air volume of the indoor fan **33a** may be controlled such that the air volume is set to one of three or more air volumes. The air volume of the indoor fan **33a** is controlled depending on the value of the sensible heat ratio difference ΔSHF and reference values $X11$ to $X13$ ($X11 > X12 > X13$). For example, in the case where the sensible heat ratio difference ΔSHF is a small difference, the air volume of the indoor fan **33a** is controlled such that the air volume is increased. In the case where the sensible heat ratio difference ΔSHF is a large difference, the air volume of the indoor fan **33a** is controlled such that the air volume is decreased. In the first dehumidifying operation mode, the air volume of the indoor fan **33a** is thus increased when the value of the sensible heat ratio difference ΔSHF decreases. This enables the indoor space to be dehumidified because the evaporating temperature of the refrigerant that flows through the indoor heat exchanger **31a** decreases to a temperature less than or equal to the dew point of the indoor air. For example, $X13 = 0.2$ to 0.3 , $X12 = X13 + \alpha$, and $X11 = X12 + a$ are satisfied. For example, α is 0.1 . When α is less than 0.05 , however, the air volume of the indoor fan **33a** can be more finely controlled.

(53) [Second Dehumidifying Operation Mode]

(54) FIG. 4 is a chart illustrating an example of a control operation of the air-conditioning apparatus **100** according to Embodiment 1 in the second dehumidifying operation mode. In FIG. 4, the horizontal axis represents the sensible heat ratio difference, and the vertical axis represents a ratio between an ON time and an OFF time. Arrows in FIG. 4 represent transition of the sensible heat ratio difference.

(55) The second dehumidifying operation mode of the air-conditioning apparatus **100** according to Embodiment 1 will be described below with reference to FIG. 4.

(56) In the second dehumidifying operation mode, an ON mode in which the rotation frequency of the compressor **11** is controlled for the ON time $\Delta t1$ alternates with an OFF mode in which the operation of the compressor **11** is suspended for the OFF time $\Delta t2$. In the ON mode, the rotation frequency of the compressor **11** is increased to a rotation frequency higher than the rotation frequency in the cooling operation mode, and cooling capacity is increased. The flow switching device **14**, the outdoor fan **121**, and the expansion device **13** operate in the same manner as in the cooling operation mode described above. At this time, the air volume of the indoor fan **33a** is controlled by the controller **50** such that the air volume is set to the maximum possible air volume. In the OFF mode, the compressor **11** is suspended, the opening degree of the expansion device **13** is highest, and the outdoor fan **121** is suspended. At this time, the indoor fan **33a** may be suspended. The ratio between the ON time $\Delta t1$ and the OFF time $\Delta t2$ is controlled depending on the value of the sensible heat ratio difference ΔSHF and reference values $X21$ to $X23$

($X21 > X22 > X23$). For example, in the case where the sensible heat ratio difference ΔSHF is a small difference, the ratio between the ON time $\Delta t1$ and the OFF time $\Delta t2$ is controlled such that the time of the ON mode is relatively decreased. In the case where the sensible heat ratio difference ΔSHF is a large difference, the ratio between the ON time $\Delta t1$ and the OFF time $\Delta t2$ is controlled such that the time of the ON mode is relatively increased. In the second dehumidifying operation mode, the ON mode thus alternates with the OFF mode. In the ON mode, the rotation frequency of the compressor **11** is increased, and the cooling capacity is increased. This enables the amount of dehumidification to be ensured and enables the indoor space to be sufficiently dehumidified. In addition, in the OFF mode, the compressor **11** is suspended, and the indoor space can be consequently prevented from being cooled to too low temperature. For example, $X23 = 0.1$ to 0.2 , $X22 = X23 + \beta$, and $X21 = X22 + \beta$ are satisfied. For example, β is 0.1 . When β is less than 0.05 ,

however, the ratio between the ON time $\Delta t1$ and the OFF time $\Delta t2$ can be more finely controlled.

(57) The air-conditioning apparatus **100** according to Embodiment 1 described above includes the indoor unit **3a**, which is installed in the indoor space and includes the indoor heat exchanger **31a**, the indoor fan **33a**, which sends the indoor air to the indoor heat exchanger **31a**, and the indoor temperature sensor **35a**, which detects the indoor temperature T_a , the outdoor unit **1**, which is installed outside the indoor space described above and includes the compressor **11** and the outdoor heat exchanger **12**, the refrigerant circuit in which the compressor **11**, the outdoor heat exchanger **12**, the expansion device **13**, and the indoor heat exchanger **31a** are connected by the pipes, and the controller **50** having the cooling operation mode in which the indoor space is cooled and the multiple dehumidifying operation modes in which the indoor space is dehumidified. The multiple dehumidifying operation modes include the first dehumidifying operation mode in which the air volume of the indoor fan **33a** is increased or decreased and the second dehumidifying operation mode in which the compressor **11** alternates between operation and suspension. When the cooling operation mode is changed into one of the dehumidifying operation modes, the controller **50** selects the one of the multiple dehumidifying operation modes on the basis of the value of the sensible heat ratio difference ΔSHF , which is a difference between the target sensible heat ratio SHF acquired from the indoor temperature T_a detected by the indoor temperature sensor **35a**, the target indoor temperature T_m , and the target indoor humidity and the theoretical minimum possible sensible heat ratio SHF_m acquired from the enthalpy h_i of the indoor air, the enthalpy h_o of the blown air from the indoor unit **3a** when the relative humidity is 100%, and the enthalpy Δh_s of the sensible heat.

(58) In the air-conditioning apparatus **100** according to Embodiment 1, the controller **50** selects one of the multiple dehumidifying operation modes having different dehumidifying capacities on the basis of the value of the sensible heat ratio difference ΔSHF , which is a difference between the target sensible heat ratio SHF and the theoretical minimum possible sensible heat ratio SHF_m when the cooling operation mode is changed into the one of the dehumidifying operation modes. This enables the optimal possible dehumidifying operation mode for dehumidification to be selected, prevents the evaporating temperature from greatly decreasing, and enables dehumidification to be performed without reducing user comfort.

(59) In the air-conditioning apparatus **100** according to Embodiment 1, the controller **50** changes the cooling operation mode into the first dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is larger than a first reference value. The controller **50** changes the cooling operation mode into the second dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is not larger than the first reference value.

(60) In the air-conditioning apparatus **100** according to Embodiment 1, the controller **50** makes a change into the first dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is larger than the first reference value and makes a change into the second dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is not larger than the first reference value. That is, when the sensible heat ratio difference ΔSHF is a large difference, and there is no possibility that the indoor space is cooled to too low temperature, the controller **50** makes the change into the first dehumidifying operation mode in which the evaporating temperature is controlled by controlling the air volume of the indoor fan **33a**. When the sensible heat ratio difference ΔSHF is a small difference, and there is a possibility that the indoor space is cooled to too low temperature, the controller **50** makes the change into the second dehumidifying operation mode in which the evaporating temperature is controlled by controlling the on-off and rotation frequency of the compressor **11**. This enables the optimal possible dehumidifying operation mode for dehumidification to be selected, prevents the evaporating temperature from greatly decreasing, and enables dehumidification to be performed without reducing the user comfort.

(61) In the air-conditioning apparatus **100** according to Embodiment 1, the controller **50** determines

the air volume of the indoor fan **33a** on the basis of the value of the sensible heat ratio difference ΔSHF in the first dehumidifying operation mode. The controller **50** then decreases the air volume of the indoor fan **33a** when the value of the sensible heat ratio difference ΔSHF decreases.

(62) In the air-conditioning apparatus **100** according to Embodiment 1, the air volume of the indoor fan **33a** is decreased when the value of the sensible heat ratio difference ΔSHF decreases.

Consequently, the evaporating temperature of the refrigerant that flows through the indoor heat exchanger **31a** decreases to a temperature less than or equal to the dew point of the indoor air, and the indoor space can be dehumidified.

(63) In the air-conditioning apparatus **100** according to Embodiment 1, in the second dehumidifying operation mode, the controller **50** alternates between the ON mode and the OFF mode, the rotation frequency of the compressor **11** is controlled for the ON time $\Delta t1$ such that the indoor temperature Ta detected by the indoor temperature sensor **35a** approaches the target indoor temperature Tm in the ON mode, and the operation of the compressor **11** is suspended for the OFF time $\Delta t2$ in the OFF mode. The controller **50** then determines the ratio between the ON time $\Delta t1$ and the OFF time $\Delta t2$ on the basis of the value of the sensible heat ratio difference ΔSHF .

(64) The air-conditioning apparatus **100** according to Embodiment 1 alternates between the ON mode and the OFF mode and can consequently prevent the indoor space from being cooled to too low temperature while sufficiently dehumidifying the indoor space.

Embodiment 2

(65) Embodiment 2 will be described below, where a description for a content common to Embodiment 1 is omitted, and components that are the same as or correspond to the components in Embodiment 1 are designated by the same reference signs.

(66) [Air-Conditioning Apparatus]

(67) FIG. 5 is a refrigerant circuit diagram illustrating an example of the structure of an air-conditioning apparatus **100** according to Embodiment 2.

(68) As illustrated in FIG. 5, the air-conditioning apparatus **100** according to Embodiment 2 includes the outdoor unit **1** and the indoor unit **3a**. The outdoor unit **1** and the indoor unit **3a** are connected to each other by the gas pipe **15** and the liquid pipe **16** and are included in the refrigerant circuit.

(69) [Indoor Unit]

(70) The indoor unit **3a** is installed, for example, inside a room and supplies air for air conditioning to an indoor space. The indoor unit **3a** includes the indoor heat exchanger **31a**, an auxiliary heat exchanger **32a**, and an indoor expansion device **40**. The indoor heat exchanger **31a** and the auxiliary heat exchanger **32a** are connected to each other by a pipe and across the indoor expansion device **40**. The indoor unit **3a** also includes the indoor fan **33a**, which sends air to the indoor heat exchanger **31a** and the auxiliary heat exchanger **32a**.

(71) The indoor heat exchanger **31a** and the auxiliary heat exchanger **32a** exchange heat between the air and refrigerant, consequently condenses or evaporates the refrigerant, and generates heating air or cooling air to be supplied to the indoor space. The indoor fan **33a** sends indoor air to the indoor heat exchanger **31a** and the auxiliary heat exchanger **32a**. The indoor fan **33a** is driven by the indoor fan motor **34a**. The indoor fan **33a** is disposed at an air passage upstream of the indoor heat exchanger **31a** and the auxiliary heat exchanger **32a** but is not limited to such a location. The indoor fan **33a** may be disposed at the air passage downstream of the indoor heat exchanger **31a** and the auxiliary heat exchanger **32a**. The indoor expansion device **40** decompresses and expands the refrigerant and is, for example, an on-off valve that opens to a fixed opening degree or closes.

(72) The indoor unit **3a** also includes the indoor temperature sensor **35a**, the entrance temperature sensor **36a**, the exit temperature sensor **37a**, and an intermediate temperature sensor **38a**. The indoor temperature sensor **35a** is, for example, a thermistor and detects the indoor temperature Ta . The indoor temperature sensor **35a** is disposed at an air inlet of the indoor unit **3a** for the indoor air during a cooling operation. The entrance temperature sensor **36a** is, for example, a thermistor and

detects the temperature of the refrigerant that flows into the auxiliary heat exchanger **32a** during the cooling operation. The entrance temperature sensor **36a** is disposed at the pipe at the entrance of the auxiliary heat exchanger **32a** for the refrigerant during the cooling operation. The exit temperature sensor **37a** is, for example, a thermistor and detects the temperature of the refrigerant that flows out from the indoor heat exchanger **31a** during the cooling operation. The exit temperature sensor **37a** is disposed at the exit of the indoor heat exchanger **31a** for the refrigerant during the cooling operation. The intermediate temperature sensor **38a** is, for example, a thermistor and detects the temperature of the refrigerant that flows into the indoor heat exchanger **31a** during the cooling operation. The intermediate temperature sensor **38a** is disposed at the pipe between the indoor expansion device **40** and the indoor heat exchanger **31a**.

(73) The structure of the outdoor unit **1** according to Embodiment 2 is the same as the structure according to Embodiment 1 and is thus not described.

(74) The refrigerant circuit of the air-conditioning apparatus **100** according to Embodiment 2 is formed by connecting the compressor **11**, the flow switching device **14**, the outdoor heat exchanger **12**, the expansion device **13**, the auxiliary heat exchanger **32a**, the indoor expansion device **40**, and the indoor heat exchanger **31a** in this order by the pipes.

(75) [Cooling Operation Mode]

(76) The cooling operation mode in which the indoor heat exchanger **31a** of the air-conditioning apparatus **100** according to Embodiment 2 generates a cooling load will be described.

(77) The flow of the refrigerant will first be described. The compressor **11** sucks and compresses low-temperature and low-pressure gas refrigerant and discharges high-temperature and high-pressure gas refrigerant. The high-temperature and high-pressure gas refrigerant that is discharged from the compressor **11** flows into the outdoor heat exchanger **12** via the flow switching device **14**. The outdoor heat exchanger **12** exchanges heat between the outdoor air that is supplied from the outdoor fan **121** and the high-temperature and high-pressure gas refrigerant. Intermediate-temperature and high-pressure refrigerant that is acquired by cooling the refrigerant at the outdoor heat exchanger **12** flows into the expansion device **13**. The expansion device **13** decompresses the intermediate-temperature and high-pressure refrigerant in a two-phase or liquid state. Low-temperature and low-pressure two-phase refrigerant that is acquired by decompressing the refrigerant at the expansion device **13** flows into the auxiliary heat exchanger **32a** via the liquid pipe **16**. At this time, the indoor unit **3a** performs the cooling operation, and the auxiliary heat exchanger **32a** exchanges heat between the indoor air that is supplied from the indoor fan **33a** and the low-temperature refrigerant. The refrigerant that is heated by the auxiliary heat exchanger **32a** flows into the indoor heat exchanger **31a** via the indoor expansion device **40**, which fully opens. At this time, the indoor unit **3a** performs the cooling operation, and the indoor heat exchanger **31a** exchanges heat between the indoor air that is supplied from the indoor fan **33a** and the refrigerant. Intermediate-temperature refrigerant that is acquired by heating the refrigerant at the indoor heat exchanger **31a** passes through the gas pipe **15** and the flow switching device **14** and is sucked by the compressor **11**.

(78) The compressor **11** is controlled such that the temperature difference ΔT between the target indoor temperature T_m and the indoor temperature T_a detected by the indoor temperature sensor **35a** approaches 0. For example, the larger the temperature difference ΔT , the higher the rotation frequency of the compressor **11**, and the smaller the temperature difference ΔT , the lower the rotation frequency of the compressor **11**. An air volume from the indoor unit **3a** is controlled by the controller **50**. The controller **50** controls the air volume from the indoor unit **3a** within an adjustable range. For example, the upper limit of the adjustable range is 100% of the maximum possible air volume of the indoor fan **33a**, and the lower limit of the adjustable range is 70% of the maximum possible air volume.

(79) [Determination of Dehumidifying Operation Mode]

(80) FIG. 6 is a flowchart for the air-conditioning apparatus **100** according to Embodiment 2 when

the cooling operation mode is changed into one of dehumidifying operation modes.

(81) The dehumidifying operation modes of the air-conditioning apparatus **100** according to Embodiment 2 include the first dehumidifying operation mode, the second dehumidifying operation mode, and a third dehumidifying operation mode.

(82) The flow of change from the cooling operation mode into one of the dehumidifying operation modes of the air-conditioning apparatus **100** according to Embodiment 2 will be described below with reference to FIG. 6.

(83) (Step S11)

(84) The controller **50** acquires the indoor temperature T_a detected by the indoor temperature sensor **35a**. Subsequently, processing proceeds to step S12.

(85) (Step S12)

(86) The controller **50** acquires the target sensible heat ratio SHF from the acquired indoor temperature T_a , the target indoor temperature T_m , and the target indoor humidity. Subsequently, the processing proceeds to step S13.

(87) (Step S13)

(88) The controller **50** acquires the sensible heat ratio difference ΔSHF , which is a difference between the theoretical minimum possible sensible heat ratio SHF_m and the target sensible heat ratio SHF. Subsequently, the processing proceeds to step S14.

(89) The theoretical minimum possible sensible heat ratio SHF_m is acquired by the expression described in Embodiment 1.

(90) (Step S14)

(91) The controller **50** determines whether the sensible heat ratio difference ΔSHF is larger than the reference value $X1$. When the controller **50** determines that the sensible heat ratio difference ΔSHF is larger than the reference value $X1$ (YES), the processing proceeds to step S16. On the other hand, when the controller **50** determines that the sensible heat ratio difference ΔSHF is not larger than the reference value $X1$ (NO), the processing proceeds to step S15.

(92) (Step S15)

(93) The controller **50** determines whether the sensible heat ratio difference ΔSHF is larger than a reference value $X2$ ($< X1$). When the controller **50** determines that the sensible heat ratio difference ΔSHF is larger than the reference value $X2$ (YES), the processing proceeds to step S17. On the other hand, when the controller **50** determines that the sensible heat ratio difference ΔSHF is not larger than the reference value $X2$ (NO), the processing proceeds to step S18. The reference value $X2$ is, for example, 0.1 to 0.2, is determined depending on the dew point of the indoor air, and increases as the dew point of the indoor air decreases.

(94) (Step S16)

(95) The controller **50** changes the operation mode from the cooling operation mode into the first dehumidifying operation mode. Subsequently, the processing returns to step S11.

(96) (Step S17)

(97) The controller **50** changes the operation mode from the cooling operation mode into the second dehumidifying operation mode. Subsequently, the processing returns to step S11.

(98) (Step S18)

(99) The controller **50** changes the operation mode from the cooling operation mode into the third dehumidifying operation mode. Subsequently, the processing returns to step S11.

(100) [Third Dehumidifying Operation Mode]

(101) The third dehumidifying operation mode of the air-conditioning apparatus **100** according to Embodiment 2 will be described below. The first dehumidifying operation mode and the second dehumidifying operation mode of the air-conditioning apparatus **100** according to Embodiment 2 are the same as those according to Embodiment 1 and are not described.

(102) The compressor **11** sucks and compresses low-temperature and low-pressure gas refrigerant and discharges high-temperature and high-pressure gas refrigerant. The high-temperature and high-

pressure gas refrigerant that is discharged from the compressor **11** flows into the outdoor heat exchanger **12** via the flow switching device **14**. The outdoor heat exchanger **12** exchanges heat between the outdoor air that is supplied from the outdoor fan **121** and the high-temperature and high-pressure gas refrigerant. Intermediate-temperature and high-pressure refrigerant that is acquired by cooling the refrigerant at the outdoor heat exchanger **12** flows into the auxiliary heat exchanger **32a** via the expansion device **13**. At this time, the opening degree of the expansion device **13** is highest. The auxiliary heat exchanger **32a** exchanges heat between the indoor air that is supplied from the indoor fan **33a** and the high-temperature and high-pressure refrigerant. The indoor expansion device **40** decompresses intermediate-temperature refrigerant that is acquired by cooling the refrigerant at the auxiliary heat exchanger **32a** into low-temperature and low-pressure refrigerant, and the low-temperature and low-pressure refrigerant flows into the indoor heat exchanger **31a**. At this time, the indoor heat exchanger **31a** exchanges heat between the indoor air that is supplied from the indoor fan **33a** and the low-temperature and low-pressure refrigerant. Intermediate-temperature and low-pressure refrigerant that is acquired by heating the refrigerant at the indoor heat exchanger **31a** passes through the gas pipe **15** and the flow switching device **14** and is sucked by the compressor **11**. The indoor expansion device **40** is opened or closed on the basis of the indoor temperature T_a detected by the indoor temperature sensor **35a**.

(103) FIG. 7 is a saturated air psychrometric chart illustrating an example of the state of air in the air-conditioning apparatus **100** according to Embodiment 2 in the third dehumidifying operation mode. In FIG. 7, the horizontal axis represents the temperature of the air, the vertical axis represents absolute humidity, and a curve represents a saturated air curve.

(104) In the third dehumidifying operation mode, as illustrated in FIG. 7, the indoor air (RA in the figure) that is supplied from the indoor fan **33a** is cooled by the indoor heat exchanger **31a** (CA in the figure) and is heated by the auxiliary heat exchanger **32a** (HA in the figure). Low-temperature and low-humidity air that is acquired by cooling the indoor air by use of the indoor heat exchanger **31a** and high-temperature and intermediate-humidity air that is acquired by heating the indoor air by use of the auxiliary heat exchanger **32a** are mixed in the indoor unit **3a**, and the mixed air is subsequently supplied to the indoor space (SA in the figure). This enables intermediate-temperature and low-humidity air to be supplied from the indoor unit **3a** to the indoor space. That is, a reheat dehumidifying operation can be performed in the dehumidifying operation mode.

(105) In the air-conditioning apparatus **100** according to Embodiment 2 described above, the indoor unit **3a** includes the auxiliary heat exchanger **32a** disposed between the indoor heat exchanger **31a** and the expansion device **13** and the indoor expansion device **40** disposed between the indoor heat exchanger **31a** and the auxiliary heat exchanger **32a**. In addition, the dehumidifying operation modes include the third dehumidifying operation mode in which the indoor expansion device **40** is opened or closed on the basis of the indoor temperature T_a detected by the indoor temperature sensor **35a**.

(106) The dehumidifying operation modes of the air-conditioning apparatus **100** according to Embodiment 2 include the third dehumidifying operation mode in which the indoor expansion device **40** is opened or closed on the basis of the indoor temperature T_a detected by the indoor temperature sensor **35a**. In the third dehumidifying operation mode, the low-temperature and low-humidity air that is acquired by cooling the indoor air by use of the indoor heat exchanger **31a** and the high-temperature and intermediate-humidity air that is acquired by heating the indoor air by use of the auxiliary heat exchanger **32a** are mixed in the indoor unit **3a**, and the mixed air is subsequently supplied to the indoor space. For this reason, the intermediate-temperature and low-humidity air can be supplied from the indoor unit **3a** to the indoor space.

(107) In the air-conditioning apparatus **100** according to Embodiment 2, the controller **50** changes the cooling operation mode into the first dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is larger than the first reference value. The controller **50** changes the cooling operation mode into the second dehumidifying operation mode in the case

where the sensible heat ratio difference ΔSHF is not larger than the first reference value and is larger than the second reference value, which is a value smaller than the first reference value. The controller **50** changes the cooling operation mode into the third dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is not larger than the second reference value.

(108) In the air-conditioning apparatus **100** according to Embodiment 2, the controller **50** makes a change into the first dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is larger than the first reference value. The controller **50** makes a change into the second dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is not larger than the first reference value and is larger than the second reference value and makes a change into the third dehumidifying operation mode in the case where the sensible heat ratio difference ΔSHF is not larger than the second reference value. That is, when the sensible heat ratio difference ΔSHF is a large difference, and there is no possibility that the indoor space is cooled to too low temperature, the controller **50** makes the change into the first dehumidifying operation mode in which the evaporating temperature is controlled by controlling the air volume of the indoor fan **33a**. When the sensible heat ratio difference ΔSHF is a small difference, and there is a possibility that the indoor space is cooled to too low temperature, the controller **50** makes the change into the second dehumidifying operation mode in which the evaporating temperature is controlled by controlling the on-off and rotation frequency of the compressor **11**. When the sensible heat ratio difference ΔSHF is smaller than the small difference, and there is a possibility that the indoor space is cooled to a temperature lower than the too low temperature, the controller **50** makes the change into the third dehumidifying operation mode in which the intermediate-temperature and low-humidity air is supplied from the indoor unit **3a** to the indoor space. This enables the optimal possible dehumidifying operation mode for dehumidification to be selected, prevents the evaporating temperature from greatly decreasing, and enables dehumidification to be performed without reducing the user comfort.

Embodiment 3

(109) Embodiment 3 will be described below, where a description for a content common to Embodiment 1 and Embodiment 2 is omitted, and components that are the same as or correspond to the components in Embodiment 1 and Embodiment 2 are designated by the same reference signs.

(110) [Air-Conditioning Apparatus]

(111) FIG. **8** is a refrigerant circuit diagram illustrating an example of the structure of an air-conditioning apparatus **100** according to Embodiment 3.

(112) As illustrated in FIG. **8**, the air-conditioning apparatus **100** according to Embodiment 3 includes the outdoor unit **1** and indoor units **3a** and **3b**. The outdoor unit **1** and the indoor units **3a** and **3b** are connected to each other by the gas pipe **15** and the liquid pipe **16** and are included in the refrigerant circuit. In an example illustrated in FIG. **8**, the two indoor units **3a** and **3b** are connected to the outdoor unit **1**. However, this is not a limitation, and the number of the indoor units may be three or more.

(113) [Indoor Unit]

(114) The indoor units **3a** and **3b** are installed, for example, inside a room and supply air for air conditioning to an indoor space. The indoor unit **3a** includes the indoor heat exchanger **31a**, and the indoor unit **3b** includes an indoor heat exchanger **31b**. The indoor units **3a** and **3b** are connected to the outdoor unit **1** such that the indoor units **3a** and **3b** are in parallel with each other. The indoor unit **3a** also includes the indoor fan **33a**, which sends air to the indoor heat exchanger **31a**, and the indoor unit **3b** also includes an indoor fan **33b**, which sends air to the indoor heat exchanger **31b**. The indoor fan **33a** is driven by the indoor fan motor **34a**, and the indoor fan **33b** is driven by an indoor fan motor **34b**.

(115) The indoor unit **3a** also includes the indoor temperature sensor **35a**, the entrance temperature sensor **36a**, and the exit temperature sensor **37a**, and the indoor unit **3b** also includes an indoor

temperature sensor **35b**, an entrance temperature sensor **36b**, and an exit temperature sensor **37b**. The indoor temperature sensor **35a** is, for example, a thermistor and detects the indoor temperature T_a , and the indoor temperature sensor **35b** is, for example, a thermistor and detects an indoor temperature T_b . The indoor temperature sensor **35a** is disposed at an air inlet of the indoor unit **3a** for indoor air, and the indoor temperature sensor **35b** is disposed at an air inlet of the indoor unit **3b** for the indoor air. The entrance temperature sensor **36a** is, for example, a thermistor and detects the temperature of refrigerant that flows into the indoor heat exchanger **31a** during a cooling operation, and the entrance temperature sensor **36b** is, for example, a thermistor and detects the temperature of refrigerant that flows into the indoor heat exchanger **31b** during the cooling operation. The entrance temperature sensor **36a** is disposed at a pipe at the entrance of the indoor heat exchanger **31a** for the refrigerant during the cooling operation, and the entrance temperature sensor **36b** is disposed at a pipe at the entrance of the indoor heat exchanger **31b** for the refrigerant during the cooling operation. The exit temperature sensor **37a** is, for example, a thermistor and detects the temperature of the refrigerant that flows out from the indoor heat exchanger **31a** during the cooling operation, and the exit temperature sensor **37b** is, for example, a thermistor and detects the temperature of the refrigerant that flows out from the indoor heat exchanger **31b** during the cooling operation. The exit temperature sensor **37a** is disposed at the exit of the indoor heat exchanger **31a** for the refrigerant during the cooling operation, and the exit temperature sensor **37b** is disposed at the exit of the indoor heat exchanger **31b** for the refrigerant during the cooling operation.

(116) The refrigerant circuit of the air-conditioning apparatus **100** according to Embodiment 3 is formed by connecting the compressor **11**, the flow switching device **14**, the outdoor heat exchanger **12**, the expansion device **13**, and the indoor heat exchangers **31a** and **31b** in this order by the pipes.

(117) The structure of the outdoor unit **1** according to Embodiment 3 is the same as the structure according to Embodiment 1 and is thus not described.

(118) [Cooling Operation Mode]

(119) The cooling operation mode of the air-conditioning apparatus **100** according to Embodiment 3 in which the indoor heat exchangers **31a** and **31b** generate a cooling load will be described.

(120) The flow of the refrigerant will first be described. The compressor **11** sucks and compresses low-temperature and low-pressure gas refrigerant and discharges high-temperature and high-pressure gas refrigerant. The high-temperature and high-pressure gas refrigerant that is discharged from the compressor **11** flows into the outdoor heat exchanger **12** via the flow switching device **14**. The outdoor heat exchanger **12** exchanges heat between the outdoor air that is supplied from the outdoor fan **121** and the high-temperature and high-pressure gas refrigerant. Intermediate-temperature and high-pressure refrigerant that is acquired by cooling the refrigerant at the outdoor heat exchanger **12** flows into the expansion device **13**. The expansion device **13** decompresses the intermediate-temperature and high-pressure refrigerant in a two-phase or liquid state. Low-temperature and low-pressure two-phase refrigerant that is acquired by decompressing the refrigerant at the expansion device **13** flows into the indoor heat exchangers **31a** and **31b** via the liquid pipe **16**. At this time, the indoor units **3a** and **3b** each perform the cooling operation, the indoor heat exchanger **31a** exchanges heat between the indoor air that is supplied from the indoor fan **33a** and the low-temperature refrigerant, and the indoor heat exchanger **31b** exchanges heat between the indoor air that is supplied from the indoor fan **33b** and the low-temperature refrigerant. Intermediate-temperature refrigerant that is acquired by heating the refrigerant at the indoor heat exchangers **31a** and **31b** joins with each other in the gas pipe **15**, passes through the gas pipe **15** and the flow switching device **14**, and is sucked by the compressor **11**.

(121) The compressor **11** is controlled such that an average temperature difference ΔT_{ave} acquired by averaging a temperature difference ΔT_1 between the target indoor temperature T_m and the indoor temperature T_a detected by the indoor temperature sensor **35a** and a temperature difference ΔT_2 between the target indoor temperature T_m and the indoor temperature T_b detected by the indoor temperature sensor **35b** approaches 0. For example, the larger the average temperature

difference ΔT_{ave} , the higher the rotation frequency of the compressor **11**, and the smaller the average temperature difference ΔT_{ave} , the lower the rotation frequency of the compressor **11**. Air volumes from the indoor units **3a** and **3b** are controlled by the controller **50**. The controller **50** controls the air volumes from the indoor units **3a** and **3b** within adjustable ranges. For example, the upper limits of the adjustable ranges are 100% of the respective maximum possible air volumes of the indoor fans **33a** and **33b**, and the lower limits of the adjustable ranges are 70% of the respective maximum possible air volumes.

(122) [Conditions for Change into Dehumidifying Operation Modes]

(123) Conditions for change from the cooling operation mode into dehumidifying operation modes will be described below. When the average temperature difference ΔT_{ave} described above becomes a predetermined temperature or less, and the amount of change in the rotation frequency of the compressor **11** becomes a predetermined value or less, whether a change into one of the dehumidifying operation modes is made is determined.

(124) [Determination of Dehumidifying Operation Mode]

(125) FIG. **9** is a flowchart for the air-conditioning apparatus **100** according to Embodiment 3 when the cooling operation mode is changed into one of the dehumidifying operation modes.

(126) The dehumidifying operation modes of the air-conditioning apparatus **100** according to Embodiment 3 include the first dehumidifying operation mode and the second dehumidifying operation mode.

(127) The flow of change from the cooling operation mode into one of the dehumidifying operation modes of the air-conditioning apparatus **100** according to Embodiment 3 will be described below with reference to FIG. **9**.

(128) (Step S21)

(129) The controller **50** acquires the indoor temperature T_a detected by the indoor temperature sensor **35a** and the indoor temperature T_b detected by the indoor temperature sensor **35b**. Subsequently, processing proceeds to step S22.

(130) (Step S22)

(131) The controller **50** acquires the target sensible heat ratio SHF from an average indoor temperature T_{ave} acquired by averaging the indoor temperature T_a detected by the indoor temperature sensor **35a** and the indoor temperature T_b detected by the indoor temperature sensor **35b**, the target indoor temperature T_m , and the target indoor humidity. Subsequently, the processing proceeds to step S23.

(132) (Step S23)

(133) The controller **50** acquires the sensible heat ratio difference ΔSHF , which is a difference between the theoretical minimum possible sensible heat ratio SHF_m and the target sensible heat ratio SHF. Subsequently, the processing proceeds to step S24. The theoretical minimum possible sensible heat ratio SHF_m is acquired by the expression described in Embodiment 1.

(134) (Step S24)

(135) The controller **50** determines whether the sensible heat ratio difference ΔSHF is larger than the reference value $X1$. When the controller **50** determines that the sensible heat ratio difference ΔSHF is larger than the reference value $X1$ (YES), the processing proceeds to step S25. On the other hand, when the controller **50** determines that the sensible heat ratio difference ΔSHF is not larger than the reference value $X1$ (NO), the processing proceeds to step S26.

(136) (Step S25)

(137) The controller **50** changes the operation mode from the cooling operation mode into the first dehumidifying operation mode. Subsequently, the processing returns to step S21.

(138) (Step S26)

(139) The controller **50** changes the operation mode from the cooling operation mode into the second dehumidifying operation mode. Subsequently, the processing returns to step S21.

(140) For the processing described above by way of example, the multiple indoor units **3a** and **3b**

are installed in the same indoor space, and the target indoor temperature and the target indoor humidity are shared. Even in the case where the indoor units **3a** and **3b** are installed in different indoor spaces, and the indoor spaces have different target indoor temperatures and different values of the target indoor humidity, the processing described above can be used, for example, provided that the average value of the target indoor temperatures of the indoor spaces is used as the target indoor temperatures, and the average value of the values of the target indoor humidity of the indoor spaces is used as the values of the target indoor humidity.

(141) [First Dehumidifying Operation Mode]

(142) The first dehumidifying operation mode of the air-conditioning apparatus **100** according to Embodiment 3 will be described below with reference to FIG. 3.

(143) In the first dehumidifying operation mode, the controller **50** controls the air volumes of the indoor fans **33a** and **33b**. The lower limits of the air volumes of the indoor fans **33a** and **33b** may be smaller than the respective lower limits in the cooling operation mode. As illustrated in FIG. 3, the air volumes of the indoor fans **33a** and **33b** are each controlled depending on the magnitude of the sensible heat ratio difference ΔSHF . In FIG. 3, the air volumes of the indoor fans **33a** and **33b** are each controlled such that the air volume is set to one of the three air volumes 1 to 3 (the air volume 1 > the air volume 2 > the air volume 3). However, this is not a limitation. The air volumes of the indoor fans **33a** and **33b** may be each controlled such that the air volume is set to one of three or more air volumes. The air volumes of the indoor fans **33a** and **33b** are each controlled depending on the value of the sensible heat ratio difference ΔSHF and the reference values **X11** to **X13**. For example, in the case where the sensible heat ratio difference ΔSHF is a small difference, the air volumes of the indoor fans **33a** and **33b** are each controlled such that the air volume is decreased. In the case where the sensible heat ratio difference ΔSHF is a large difference, the air volumes of the indoor fans **33a** and **33b** are each controlled such that the air volume is increased. In the first dehumidifying operation mode, the air volumes of the indoor fans **33a** and **33b** are thus each decreased when the value of the sensible heat ratio difference ΔSHF decreases. This enables the indoor space to be dehumidified because the evaporating temperature of the refrigerant that flows through the indoor heat exchangers **31a** and **31b** decreases to a temperature less than or equal to the dew point of the indoor air.

(144) [Second Dehumidifying Operation Mode]

(145) The second dehumidifying operation mode of the air-conditioning apparatus **100** according to Embodiment 3 will be described below with reference to FIG. 4.

(146) In the second dehumidifying operation mode, the ON mode in which the rotation frequency of the compressor **11** is controlled for the ON time $\Delta t1$ alternates with the OFF mode in which the operation of the compressor **11** is suspended for the OFF time $\Delta t2$. In the ON mode, the rotation frequency of the compressor **11** is increased to a rotation frequency higher than the rotation frequency in the cooling operation mode, and the cooling capacity is increased. The flow switching device **14**, the outdoor fan **121**, and the expansion device **13** operate in the same manner as in the cooling operation mode described above. At this time, the air volumes of the indoor fans **33a** and **33b** are each controlled by the controller **50** such that the air volume is set to the maximum possible air volume. In the OFF mode, the compressor **11** is suspended, the opening degree of the expansion device **13** is highest, and the outdoor fan **121** is suspended. At this time, the indoor fans **33a** and **33b** may be suspended. The ratio between the ON time $\Delta t1$ and the OFF time $\Delta t2$ is controlled depending on the value of the sensible heat ratio difference ΔSHF and the reference values **X21** to **X23**. For example, in the case where the sensible heat ratio difference ΔSHF is a small difference, the ratio between the ON time $\Delta t1$ and the OFF time $\Delta t2$ is controlled such that the time of the ON mode is relatively increased. In the case where the sensible heat ratio difference ΔSHF is a large difference, the ratio between the ON time $\Delta t1$ and the OFF time $\Delta t2$ is controlled such that the time of the ON mode is relatively decreased. The ON mode thus alternates with the OFF mode. In the ON mode, the rotation frequency of the compressor **11** is increased, and the cooling capacity is

increased. This enables the amount of dehumidification to be ensured and enables the indoor space to be sufficiently dehumidified. In addition, in the OFF mode, the compressor **11** is suspended, and the indoor space can be consequently prevented from being cooled to too low temperature.

(147) At step S22 in FIG. **9**, the controller **50** may acquire the target sensible heat ratio SHF from the highest indoor temperature Tmax among the indoor temperatures Ta and Tb detected by the respective indoor temperature sensors **35a** and **35b**, the target indoor temperature Tm, and the target indoor humidity. This enables target dehumidifying capacity to be increased and enables the indoor air to be sufficiently dehumidified in the case where the temperature of the air in the indoor space in which the indoor units **3a** and **3b** are installed is not uniform.

(148) At step S22 in FIG. **9**, the controller **50** may acquire the target sensible heat ratio SHF from the highest indoor temperature Tmax, the target indoor temperature Tm, and the target indoor humidity only when a difference between the highest indoor temperature Tmax and the lowest indoor temperature Tm in among the indoor temperatures Ta and Tb detected by the respective indoor temperature sensors **35a** and **35b** is more than or equal to a predetermined value, for example, 5 degrees. In the case where the temperature difference between the highest indoor temperature Tmax and the lowest indoor temperature Tm in is less than a predetermined value, for example, 5 degrees, the target sensible heat ratio SHF is acquired through a method described for step S22 in FIG. **9**. This enables dehumidifying capacity to be increased only when the temperature of the air in the indoor space in which the indoor units **3a** and **3b** are installed is not uniform to a certain extent and prevents the indoor temperature from being too low temperature.

(149) The air-conditioning apparatus **100** according to Embodiment 3, which includes the multiple indoor units **3a** and **3b**, has the same effects as in Embodiment 1, as described above.

(150) The air-conditioning apparatus **100** according to Embodiment 3 described above includes the multiple indoor units **3a** and **3b**, and the indoor units **3a** and **3b** are connected to the outdoor unit **1** by the pipes such that the indoor units **3a** and **3b** are in parallel with each other. The controller **50** then acquires the target sensible heat ratio SHF from the average indoor temperature Tave acquired by averaging the indoor temperatures Ta and Tb detected by the respective indoor temperature sensors **35a** and **35b** included in the respective indoor units **3a** and **3b**, the target indoor temperature Tm, and the target indoor humidity. Alternately, the controller **50** acquires the target sensible heat ratio SHF from the highest indoor temperature Tmax among the indoor temperatures Ta and Tb detected by the respective indoor temperature sensors **35a** and **35b** included in the respective indoor units **3a** and **3b**, the target indoor temperature Tm, and the target indoor humidity. Alternately, in the case where the temperature difference between the highest indoor temperature Tmax and the lowest indoor temperature Tm in among the indoor temperatures Ta and Tb detected by the respective indoor temperature sensors **35a** and **35b** included in the respective indoor units **3a** and **3b** is more than or equal to a predetermined value, the controller **50** acquires the target sensible heat ratio SHF from the highest indoor temperature Tmax, the target indoor temperature Tm, and the target indoor humidity.

(151) The air-conditioning apparatus **100** according to Embodiment 3, which includes the multiple indoor units **3a** and **3b**, has the same effects as in Embodiment 1.

Embodiment 4

(152) Embodiment 4 will be described below, where a description for a content common to Embodiment 1 to Embodiment 3 is omitted, and components that are the same as or correspond to the components in Embodiment 1 to Embodiment 3 are designated by the same reference signs.

(153) The structure of an air-conditioning apparatus **100** according to Embodiment 4 is the same as the structure according to Embodiment 1 illustrated in FIG. **1** and is thus not described.

(154) [Determination of Dehumidifying Operation Mode]

(155) FIG. **10** is a flowchart for the air-conditioning apparatus **100** according to Embodiment 4 when the cooling operation mode is changed into one of dehumidifying operation modes.

(156) The dehumidifying operation modes of the air-conditioning apparatus **100** according to

Embodiment 4 include the first dehumidifying operation mode and the second dehumidifying operation mode.

(157) The flow of change from the cooling operation mode into one of the dehumidifying operation modes of the air-conditioning apparatus **100** according to Embodiment 4 will be described below with reference to FIG. **10**.

(158) (Step **S31**)

(159) The controller **50** acquires the temperature of refrigerant that is detected by the entrance temperature sensor **36a** and that flows into the indoor heat exchanger **31a**. Subsequently, processing proceeds to step **S32**.

(160) (Step **S32**)

(161) The controller **50** acquires a dew point T_{dp} from the target indoor temperature T_m and the target indoor humidity. Subsequently, the processing proceeds to step **S33**.

(162) (Step **S33**)

(163) The controller **50** acquires a temperature difference ΔT_{dp} between the dew point T_{dp} and the temperature of the refrigerant that is detected by the entrance temperature sensor **36a**.

Subsequently, the processing proceeds to step **S34**.

(164) (Step **S34**)

(165) The controller **50** determines whether the temperature difference ΔT_{dp} is larger than 0. When the controller **50** determines that the temperature difference ΔT_{dp} is larger than 0 (YES), the processing proceeds to step **S36**. On the other hand, when the controller **50** determines that the temperature difference ΔT_{dp} is not larger than 0 (NO), the processing proceeds to step **S35**.

(166) (Step **S35**)

(167) The controller **50** determines whether the temperature difference ΔT_{dp} is larger than a reference value **R1** (<0). When the controller **50** determines that the temperature difference ΔT_{dp} is larger than the reference value **R1** (YES), the processing proceeds to step **S37**. On the other hand, when the controller **50** determines that the temperature difference ΔT_{dp} is not larger than the reference value **R1** (NO), the processing proceeds to step **S38**. The reference value **R1** is, for example, 4 to 10, is determined depending on the dew point T_{dp} , and decreases as the dew point T_{dp} decreases.

(168) (Step **S36**)

(169) The controller **50** continues the cooling operation mode. Subsequently, the processing returns to step **S31**.

(170) (Step **S37**)

(171) The controller **50** changes the operation mode from the cooling operation mode into the first dehumidifying operation mode. Subsequently, the processing returns to step **S31**.

(172) (Step **S38**)

(173) The controller **50** changes the operation mode from the cooling operation mode into the second dehumidifying operation mode. Subsequently, the processing returns to step **S31**.

(174) In the case where the temperature difference ΔT_{dp} is larger than 0, the cooling operation mode continues, and in the case where the temperature difference ΔT_{dp} is not larger than 0, the change into one of the dehumidifying operation modes is made as described above. Consequently, the evaporating temperature of the refrigerant that flows through the indoor heat exchanger **31a** can be less than or equal to the dew point T_{dp} of the room with certainty.

(175) The air-conditioning apparatus **100** according to Embodiment 4 described above includes the indoor unit **3a**, which is installed in the indoor space and includes the indoor heat exchanger **31a**, the indoor fan **33a**, which sends the indoor air to the indoor heat exchanger **31a**, and the entrance temperature sensor **36a**, which detects the temperature of the refrigerant that flows into the indoor heat exchanger **31a**, the outdoor unit **1**, which is installed outside the indoor space described above and includes the compressor **11** and the outdoor heat exchanger **12**, the refrigerant circuit in which the compressor **11**, the outdoor heat exchanger **12**, the expansion device **13**, and the indoor heat

exchanger **31a** are connected by the pipes, and the controller **50** having the cooling operation mode in which the indoor space is cooled and the multiple dehumidifying operation modes in which the indoor space is dehumidified. The multiple dehumidifying operation modes include the first dehumidifying operation mode in which the air volume of the indoor fan **33a** is increased or decreased and the second dehumidifying operation mode in which the compressor **11** alternates between operation and suspension. The controller **50** selects one of the multiple dehumidifying operation modes on the basis of the value of the temperature difference ΔT_{dp} between the dew point T_{dp} acquired from the target indoor temperature T_m and the target indoor humidity and the temperature of the refrigerant that is detected by the entrance temperature sensor **36a** when the cooling operation mode is changed into the one of the dehumidifying operation modes.

(176) In the air-conditioning apparatus **100** according to Embodiment 4, the controller **50** selects one of the multiple dehumidifying operation modes on the basis of the value of the temperature difference ΔT_{dp} between the dew point T_{dp} acquired from the target indoor temperature T_m and the target indoor humidity and the temperature of the refrigerant that is detected by the entrance temperature sensor **36a** when the cooling operation mode is changed into the one of the dehumidifying operation modes. This enables the optimal possible dehumidifying operation mode for dehumidification to be selected, prevents the evaporating temperature from greatly decreasing, and enables dehumidification to be performed without reducing the user comfort.

(177) In the air-conditioning apparatus **100** according to Embodiment 4, the controller **50** continues the cooling operation mode in the case where the temperature difference ΔT_{dp} is larger than 0. The controller **50** changes the cooling operation mode into the first dehumidifying operation mode in the case where the temperature difference ΔT_{dp} is larger than the reference value **R1**, which smaller than 0. The controller **50** changes the cooling operation mode into the second dehumidifying operation mode in the case where the temperature difference ΔT_{dp} is not larger than the reference value **R1**.

(178) The air-conditioning apparatus **100** according to Embodiment 4 continues the cooling operation mode in the case where the temperature difference ΔT_{dp} is larger than 0 and makes a change into one of the dehumidifying operation modes in the case where the temperature difference ΔT_{dp} is not larger than 0. This enables the evaporating temperature of the refrigerant that flows through the indoor heat exchanger **31a** to be less than or equal to the dew point T_{dp} of the room with certainty.

REFERENCE SIGNS LIST

(179) **1**: outdoor unit, **3a**: indoor unit, **3b**: indoor unit, **11**: compressor, **12**: outdoor heat exchanger, **13**: expansion device, **14**: flow switching device, **15**: gas pipe, **16**: liquid pipe, **31a**: indoor heat exchanger, **31b**: indoor heat exchanger, **32a**: auxiliary heat exchanger, **33a**: indoor fan, **33b**: indoor fan, **34a**: indoor fan motor, **34b**: indoor fan motor, **35a**: indoor temperature sensor, **35b**: indoor temperature sensor, **36a**: entrance temperature sensor, **36b**: entrance temperature sensor, **37a**: exit temperature sensor, **37b**: exit temperature sensor, **38a**: intermediate temperature sensor, **40**: indoor expansion device, **50**: controller, **100**: air-conditioning apparatus, **121**: outdoor fan, **122**: outdoor fan motor

Claims

1. An air-conditioning apparatus comprising: an indoor unit including an indoor heat exchanger, an indoor fan configured to send indoor air to the indoor heat exchanger, and an indoor temperature sensor configured to detect an indoor temperature, the indoor unit being installed in an indoor space; an outdoor unit including a compressor and an outdoor heat exchanger, the outdoor unit being installed outside the indoor space; a refrigerant circuit in which the compressor, the outdoor heat exchanger, an expansion device, and the indoor heat exchanger are connected by a pipe; and a controller having a cooling operation mode in which the indoor space is cooled and multiple

dehumidifying operation modes in which the indoor space is dehumidified, the multiple dehumidifying operation modes including a first dehumidifying operation mode in which an air volume of the indoor fan is increased or decreased and a second dehumidifying operation mode in which the compressor alternates between operation and suspension, the controller being configured to, when the cooling operation mode is changed into one of the multiple dehumidifying operation modes, select the one of the multiple dehumidifying operation modes on the basis of a value of a sensible heat ratio difference ΔSHF that is a difference between a target sensible heat ratio acquired from the indoor temperature detected by the indoor temperature sensor, a target indoor temperature, and target indoor humidity and a theoretical minimum possible sensible heat ratio acquired from enthalpy of the indoor air, enthalpy of blown air from the indoor unit when relative humidity is 100%, and enthalpy of sensible heat.

2. The air-conditioning apparatus of claim 1, wherein the controller is configured to, in the first dehumidifying operation mode, determine the air volume of the indoor fan on the basis of the value of the sensible heat ratio difference ΔSHF , and decrease the air volume of the indoor fan when the value of the sensible heat ratio difference ΔSHF decreases.

3. The air-conditioning apparatus of claim 1, wherein the controller is configured to, in the second dehumidifying operation mode, alternate between an ON mode and an OFF mode, a rotation frequency of the compressor being controlled for an ON time Δt_1 such that the indoor temperature detected by the indoor temperature sensor approaches the target indoor temperature in the ON mode, the operation of the compressor being suspended for an OFF time Δt_2 in the OFF mode, and determine a ratio between the ON time Δt_1 and the OFF time Δt_2 on the basis of the value of the sensible heat ratio difference ΔSHF .

4. The air-conditioning apparatus of claim 1, wherein the indoor unit includes an auxiliary heat exchanger disposed between the indoor heat exchanger and the expansion device, and an indoor expansion device disposed between the indoor heat exchanger and the auxiliary heat exchanger, and wherein the multiple dehumidifying operation modes include a third dehumidifying operation mode in which the indoor expansion device is opened or closed on the basis of the indoor temperature detected by the indoor temperature sensor.

5. The air-conditioning apparatus of claim 1, wherein the controller is configured to change the cooling operation mode into the first dehumidifying operation mode in a case where the sensible heat ratio difference ΔSHF is larger than a first reference value, and change the cooling operation mode into the second dehumidifying operation mode in a case where the sensible heat ratio difference ΔSHF is not larger than the first reference value.

6. The air-conditioning apparatus of claim 4, wherein the controller is configured to change the cooling operation mode into the first dehumidifying operation mode in a case where the sensible heat ratio difference ΔSHF is larger than a first reference value, change the cooling operation mode into the second dehumidifying operation mode in a case where the sensible heat ratio difference ΔSHF is not larger than the first reference value and is larger than a second reference value that is a value smaller than the first reference value, and change the cooling operation mode into the third dehumidifying operation mode in a case where the sensible heat ratio difference ΔSHF is not larger than the second reference value.

7. The air-conditioning apparatus of claim 1, wherein the air-conditioning apparatus comprises a plurality of the indoor units, and wherein the plurality of the indoor units is connected to the outdoor unit by a pipe such that the plurality of the indoor units is in parallel with each other.

8. The air-conditioning apparatus of claim 7, wherein the controller is configured to acquire the target sensible heat ratio from an average indoor temperature acquired by averaging indoor temperatures detected by the indoor temperature sensors included in the plurality of the indoor units, the target indoor temperature, and the target indoor humidity.

9. The air-conditioning apparatus of claim 7, wherein the controller is configured to acquire the target sensible heat ratio from a highest indoor temperature of indoor temperatures detected by the

indoor temperature sensors included in the plurality of the indoor units, the target indoor temperature, and the target indoor humidity.

10. The air-conditioning apparatus of claim 7, wherein the controller is configured to, in a case where a temperature difference between a highest indoor temperature and a lowest indoor temperature of indoor temperatures detected by the indoor temperature sensors included in the plurality of the indoor units is more than or equal to a predetermined value, acquire the target sensible heat ratio from the highest indoor temperature, the target indoor temperature, and the target indoor humidity.

11. An air-conditioning apparatus comprising: an indoor unit including an indoor heat exchanger, an indoor fan configured to send indoor air to the indoor heat exchanger, and an entrance temperature sensor configured to detect a temperature of refrigerant flowing into the indoor heat exchanger, the indoor unit being installed in an indoor space; an outdoor unit including a compressor and an outdoor heat exchanger, the outdoor unit being installed outside the indoor space; a refrigerant circuit in which the compressor, the outdoor heat exchanger, an expansion device, and the indoor heat exchanger are connected by a pipe; and a controller having a cooling operation mode in which the indoor space is cooled and multiple dehumidifying operation modes in which the indoor space is dehumidified, the multiple dehumidifying operation modes including a first dehumidifying operation mode in which an air volume of the indoor fan is increased or decreased and a second dehumidifying operation mode in which the compressor alternates between operation and suspension, the controller being configured to, when the cooling operation mode is changed into one of the multiple dehumidifying operation modes, select the one of the multiple dehumidifying operation modes on the basis of a value of a temperature difference ΔT_{dp} between a dew point acquired from a target indoor temperature and target indoor humidity and the temperature of the refrigerant detected by the entrance temperature sensor.

12. The air-conditioning apparatus of claim 11, wherein the controller is configured to continue the cooling operation mode in a case where the temperature difference ΔT_{dp} is larger than 0.

13. The air-conditioning apparatus of claim 12, wherein the controller is configured to change the cooling operation mode into the first dehumidifying operation mode in a case where the temperature difference ΔT_{dp} is larger than a reference value that is a value smaller than 0, and change the cooling operation mode into the second dehumidifying operation mode in a case where the temperature difference ΔT_{dp} is not larger than the reference value.
